1. Introduction

Laser-induced plasma formation (optical breakdown) offers the potential of precisely tunable nonlinear energy deposition in nominally transparent dielectrics that can be used for material processing on the micro- and nanoscale [1-10]. Laser-generated plasmas in water or aqueous fluids are of major importance for laser surgery in the eye [11-17], in other transparent tissues and cells [18-27], and for microfluidics [28-35].

Tightly focused energetic laser pulses can produce fine tunable nano-and microeffects and also generate ultra-high plasma energy densities that by far exceed the energy density in explosives such as TNT. Water is an excellent model to study the process of nonlinear energy deposition from nanoeffects to extreme states of matter because it has a large bandgap of ≈ 9.5 eV [36,37], similar to fused silica [7,38]. While the energy density required for thermoelastic bubble formation with femtosecond (fs) pulses is ≈ 0.55 kJ/cm^3 [22], the present experiments showed that under appropriate experimental conditions the plasma energy density can reach values of 260 – 400 kJ/cm^3, more than 100 times larger than the vaporization enthalpy of water. This corresponds to pressure values well above 100 GPa.

The mechanical confinement of plasmas formed in bulk water results in shock wave emission and cavitation bubble generation [39-44]. Laser generation of compact plasma with tunable energy density at well-defined time and location is an excellent tool for fundamental experimental investigations of cavitation bubble dynamics and shock wave emission in free liquid and near boundaries [40,41,45-57]. Laser-produced bubbles are not disturbed by detonation gases as in underwater explosions [58,59] or by electrodes as in spark discharge [60,61], and experimental data on their dynamics are, thus, an excellent basis for model validation [40,42,43,62-68].

In this paper, we mainly explore the optical breakdown dynamics in bulk water, with references to transparent tissues and solid dielectrics. However, the underlying mechanisms are also relevant for plasma formation at linear absorbing solid targets immersed in water, such as in laser ablation in liquids (LAL) for nanoparticle generation [69-81] and in underwater laser-induced breakdown spectroscopy (LIBS) for remote analysis of target constituents [82-89]. Here, plasma formation is ignited at the target and can, well above threshold, also spread into the liquid and shield the target. The dynamics of the bubble produced during ablation influences the particle size distribution, and during consecutive ablation events bubble remnants may disturb light transmission and ablation [74,75,90-94]. Secondary laser irradiation can be used to fragment large free nanoparticles for homogenization of the size distribution

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and modification of their crystalline structure [95-100]. In LIBS, plasma-mediated ablation is not the primary goal but a way for obtaining a characteristic spectroscopic signal from the atomized and excited target material after the initial continuum emission from the plasma has ceased [84,88,101-103]. A mechanistic understanding of optical breakdown in bulk liquid, which is the focus of this paper, also helps to understand the rules governing plasma growth and hydrodynamic phenomena in LAL and LIBS. We will use it to derive guidelines for optimizing energy coupling into the target and minimizing shielding by plasma within the liquid with the goals of optimizing ablation yield, particle properties, and spectroscopic information.

Beginning in the 1980s, laser-induced "photodisruption" has been employed for intraocular and refractive surgery, using first nanosecond (ns) pulses and later picosecond (ps) and fs pulses focused at moderate *NA* [11-13,17,47,104,105]. That stimulated research on the breakdown mechanisms at moderate numerical aperture (*NA*) with different pulse durations [12,40,47,104,106-115]. Later it was discovered that fs pulses focused at large *NA* can produce submicrometer-sized laser effects suitable for nanosurgery on a subcellular level [20-22,24,116-119] and for creating tiny voids in glass and crystalline solids [1,3,120-122]. Although laser-induced bubble formation was often employed for investigating cavitation phenomena [46,48,51,53,56,68], the process of nonlinear energy deposition itself received little attention as long as the generated bubbles had an approximately spherical shape [52,54]. Due to the difficulty of time-resolved studies on a nano- and microscale, details of the optical breakdown dynamics at large *NA* and for energies well above the breakdown threshold have experimentally been studied merely for a few individual laser parameters [19,40,110,123,124]. Attention was largely focused on the parameter dependence of the breakdown threshold and laser effects close to this threshold [36,37,112,125,126]. Furthermore, it was often restricted to ultrashort pulse durations because researchers thought that nanosecond breakdown was too stochastic and disruptive to allow for a generation of reproducible nanoeffects with precisely tunable magnitude [114,120,125,127-129].

Although optical breakdown with ultrashort laser pulses has received a lot of attention in the last three decades, various principal questions remain unresolved. In the lower range of plasma energy densities, there is a quest for compact and cost-effective devices enabling precisely adjustable nonlinear energy deposition for nanosurgery and material processing. A key question in this context is whether ns breakdown is really intrinsically stochastic and only fs laser effects are deterministic and precisely adjustable. We will show that ns breakdown is not per se stochastic, delineate the conditions for

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deterministic energy deposition with cost-effective ns pulses and demonstrate the prerequisites for producing precisely tunable nanoeffects.

An open question of great practical interest is how the partitioning of absorbed laser energy into vaporization, bubble and shock wave energy depends on plasma energy density because this partitioning governs the disruptiveness of laser effects. Of prime interest is also the question, which factors determine the upper limits of global and local plasma energy density in fs and ns breakdown and how these limits are linked to the spatiotemporal dynamics of energy deposition. Finding answers requires a precise determination of plasma size, inner structure and absorptivity.

We address the above questions through a systematic experimental exploration of the energy dependence of optical breakdown phenomena as a function of pulse duration τ_L (fs to ns), laser wavelength λ_L (UV to IR) and focusing *NA* (0.3 to 0.9). In this parameter space, we cover the range from the bubble threshold in water to extremely high plasma energy densities.

Compared to solid dielectrics, the 'self-healing' properties of water largely facilitate parametric studies. Bubble formation provides a straightforward and readily detectable breakdown criterion [37,117], and the maximum bubble radius R_{max} is a convenient measure for the magnitude of the laser effect. This way, a large amount of data can be acquired without the need for a continuous movement of the target. Tight focusing of the laser beam largely avoids nonlinear beam propagation effects [130,131], and the use of water-immersion microscope objectives guarantees aberration-free focusing [117]. A probe beam scattering technique [117] is employed to rapidly collect $R_{max}(E_L)$ data for a large number of irradiation parameters. These features are utilized to establish scaling laws for the energy dependence of nonlinear energy deposition in the (τ_L, λ_L, *NA*) parameter space.

Orthogonal adjustment of pump- and probe microscope objectives enables us to photograph plasma luminescence in side view with sub-micrometer resolution. With tight focusing, plasma luminescence can be recorded even for fs plasmas. For ns breakdown at visible wavelengths, scattered pump laser light is also recorded on the plasma colour photographs. This makes it possible to distinguish between regions of primary energy deposition and plasma inflation by radiative energy transport. Evaluation of the plasma size from the time-integrated photographs together with absorption data from transmission measurements provides values for the average plasma energy density. These data sets are then used to establish an energy balance between vaporization, shock wave and bubble energy as a function of plasma energy density. Analysis of the plasma structure and the spectral information contained in the colour

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